



LECTURE:31

Sustainable and Clean Energy



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Drivers for Renewable Energy Policies

- Altogether , these drivers might be described as the pursuit of sustainable development, where economic prosperity is advanced around the world while negative impacts are minimized.
- Renewable energy technologies provide energy services, including lighting and electricity, heating and cooling, mechanical energy and mobility.
- Relative to other types of energy (from fossil fuels, nuclear power, and traditional biomass), moderns renewables provide a variety of additional socio-economic benefits.
- These socio-economic benefits are a major force driving policymakers to adopt renewable energy targets and support policies.
- Renewable energy drivers (benefits) have been categorized in a variety of ways.



Drivers contd.....

- United Nations Intergovernmental Panel on Climate

Change (IPCC) categorized key drivers, opportunities and benefits of renewable energy into **environmental** (climate change mitigation and reduction of environmental and health impacts), energy access, energy security (e.g., diversity of fuel supply; fuel imports; balance of trade), and **social and economic development** (e.g., job creation, rural development).

Environmental Drivers-The extraction, transport, refining and use of fossil and nuclear fuels result in a host of significant **environmental impacts, including damage to land from mining; pollution of air and water**; consumption of vast amounts of fresh water, particularly for cooling at power plants; **loss of biodiversity; risk of nuclear accidents; global climate change; and associated impacts on human health**



- **World Health Organization** estimates that **outdoor air pollution – due largely to the burning of coal and road transport**. The single greatest contributor in China is pollution from coal burning (which causes an estimated 366,000 deaths annually), while in **India the major contributor is burning of solid biomass for cooking and heating**.
- **Health problems, biodiversity loss, and other environmental challenges** will only be exacerbated by climate change. **Renewable energy deployment** has become an integral part of **government strategies** around the world to address these many challenges.
- **Reduce pollution and improve public health**. Around the world, governments at all levels have **enacted policies** to support renewables in order to reduce health impacts associated with **energy production and use**.



- Concerns about the impacts of traditional use of biomass, and burning of kerosene and other fossil fuels for cooking and heating on indoor air quality, as well as the need to reduce local deforestation, also have driven **policies to promote modern renewables**.
- **Reduce fresh water use**-Many governments are turning to renewable energy to reduce water consumption associated with energy production;
- other drivers include **opportunities for local economic development and protection** against volatile fossil fuel prices.
- **Reduce reliance on nuclear power**-Germany reacted to the disaster in Japan (2011 Fukushima Daiichi nuclear disaster) by planning to phase out its own nuclear power facilities , to be replaced over time with renewable energy.
- **Mitigate climate change**- Climate change mitigation is becoming increasingly the key environmental driver of renewable energy; in combination with energy efficiency improvements, renewables now represent a key pillar in many governments' efforts to decarbonize their energy sectors.



➤ Economic Drivers

- Economic benefits, particularly for energy importers:
- 1. Renewable energy costs (especially costs of solar PV and wind power) continue their rapid decline.
- 2. Use of renewable energy helps to avoid a number of indirect economic costs associated with fossil energy production and use, such as health care expenses
- 3. It also can help reduce the longer term costs associated with global climate change, such as the potential for sudden disruption and displacement of people and their economic activity (e.g., spread of disease, forced migration).
- 4. Thus, investment in sustained economic growth is environmentally sustainable.



- **Economic benefits** associated with renewable energy that drive the adoption of support policies include:
- **1.Improve balance of trade and reduce price volatility.** The majority of countries, states or communities must import most if not all of the fossil or nuclear fuels that they consume. Investment in renewables can improve a country's or region's trade balance and can reduce fuel price volatility and supply risk.
- **For example, Denmark** expects that its strategy to move toward 100 % renewable energy (power and heat by 2035, and fossil fuel-free economy-wide by 2050), will result in reduced energy expenditures relative to business as usual.
- **2. Create jobs and develop new industries and skills** . Research suggest that while the renewable energy transition shifts jobs by sector and location, the net impact on job creation will be positive.



- Skills required for manufacturing, selling, installing and maintaining renewable energy systems and associated infrastructure vary significantly, with variety of medium- and high-skilled opportunities. Job creation has been a driver of renewable energy policies that aim to help strengthen local economies, and to stem or reverse depopulation and brain drain.
- Example : Drivers for renewable energy policies in Scotland have included diversification of rural employment and skills development. Germany to be “the first major industrialized country that achieves the transition to renewable energy with all the opportunities – for exports, development, technology, jobs – it carries with it.”
- Meet rapidly rising energy demand: The modularity of many renewable technologies and relative speed with which they can be implemented, alongside their rapidly falling costs (particularly for solar PV and wind power).



- **Provide access to energy and alleviate poverty in the Global South:** Many countries have established targets and enacted support policies to scale up renewable energy to provide access to modern energy services for people living in remote and rural areas.
- **Alleviate fuel poverty and advance rural economic development in industrialized countries:** In industrialized countries, where the vast majority of people have access to modern energy services, renewables can reduce fuel poverty and improve quality of life.
- **Keep energy revenue local:** When fuel imports are displaced with local renewables, whether at the national or sub-national level, energy expenditures can spur further economic activity in the local economy.
- **Increase tax revenue: Local** governments collect income and property tax payments from renewable energy project owners; the additional revenue enables governments to reduce tax rates for inhabitants, such as low-income residents; renewable energy projects may also reduce government expenditures.



- **Reduce public health costs:** The burning of fossil fuels for energy production results in high economic costs for societies, in addition to tremendous physical suffering . Thus it should be replaced with renewable energy in rural sector.
- **Political and Security Drivers:** Energy is critical to a healthy economy and energy services can sustain or improve people's quality of life.
- Physical availability of highly reliable and reasonably affordable energy products and services is critically important to economic and political security.
- **Improve energy security :** Renewable energy resources are diverse, they rely on natural flows ,are available locally ,and the technologies required for capturing and converting these resources into useful energy are available in the global marketplace.



- **Increase reliability and resilience** : Distributed renewable energy systems are less prone to large-scale failure; can make the power grid and other energy systems more resilient to a variety of threats, including weather-related impacts of global climate change; distributed power can be available locally (on rooftops, or from wind power projects in a city's harbor).
- **Example** : In the wake of the tsunami and nuclear disaster following **Japan's** severe earthquake in **2011**, Higashimatsushima, a city of 40,000 inhabitants, set a target for zero net energy by 2022. The city aims to transition to a safe, resilient, and sustainable energy system based on distributed renewable energy. As part of this effort, the city is building Japan's first micro-grid community (**Higashimatsushima Disaster-Prepared, Smart Eco-Town**) for evacuations in the event of future emergencies.



- **Ensure energy democracy** :The term grew out of the climate justice movement, and is grounded on the basic understanding that “the decisions that shape our lives should be established jointly and without regard to the principle of profit.
- **De-carbonization of the energy economy** is critical not only for mitigating climate change but also for achieving a more just, sustainable and resilient economy
- An **equitable , ecologically-sound energy system** should serve .the needs of the world’s peoples, and that an energy transition will be advanced by a shift to public and community control.
- **Example** : In **Japan**, as of December 2015, about **14 cities had formed companies to generate renewable energy from local resources** in order to reduce dependence on foreign energy sources.



- **Observed Impacts of Renewable Energy:**
- **Environmental**-Environmental costs of renewables are technology- and site-specific, and often can be mitigated(wildlife and humans for available land and water , and GHG emissions and other air pollutants) to some extent. Such costs need to be accounted for and in proper proportion to the environmental impacts of other energy sources.
- **Economic Impacts:** Adopting a people-centered approach and empowering citizens, farmers and small businesses to invest in renewable energy projects can be a powerful tool for socio-economic development and local wealth creation.
- **Political and Security Impacts** : **Example**-Increased deployment of renewable energy has been shown to improve energy security in the EU, where renewable's substitution for natural gas accounted for 30 % of the estimated reduction in fossil fuel demand during 2013.





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TABLE 1. REPRESENTATION OF AGGREGATE WELFARE CALCULATION
RELATIVE TO REFERENCE CASE IN 2030 (IRENA 2016)

Dimension	Welfare Function Component	Weight	Projected Change (%)	Weighted Value
Economic	Consumption and Investment	1/3	+ 0.7	0.233
Social	Employment	1/6	+ 0.1	0.017
Social	Health and Education	1/6	+ 1.8	0.300
Environmental	GHG Emissions	1/6	- 11.2	1.867
Environmental	Material Consumption	1/6	- 1.6	0.267

Total implied percentage increase in aggregate global welfare (sum of weighted): 2.68 %



- **Hydrogen as a Clean and Sustainable Energy**
- Hydrogen is also recognized as a promising and attractive energy carrier to **decarbonize the sectors responsible for global warming**, such as electricity production, industry, and transportation.
- Although hydrogen releases only water as a result of its reaction with oxygen through a fuel cell, the **hydrogen production pathway is currently a challenging issue since hydrogen is produced mainly from thermochemical processes** (natural gas reforming, coal gasification).
- On the other hand, **hydrogen production through water electrolysis** has attracted a lot of attention as a means **to reduce greenhouse gas emissions** by **using low-carbon sources** such as renewable **energy(solar, wind, hydro)** and nuclear energy.
- By providing an environmentally-friendly fuel instead of the currently-used fuels (unleaded petrol, gasoline, kerosene), **hydrogen can be used in various applications such as transportation** (aircraft, boat, vehicle, and train), **energy storage , industry, medicine, and power-to-gas.**



Addressing Challenges With Positive Steps:

- **By 2030, quintuple clean energy and climate-related R&D:** Direct public investment in research and development In total, government funding for clean energy R&D corresponds to around \$22 billion every year, or about 0.02 percent of world GDP.
- **Increased stakes in high-risk, high-reward research and development ventures**
- **translational research**
- **Govt should be flagbearer:** early-stage innovation
- **Use procurement power**
- **Tax incentives, loan guarantees**
- **Infrastructure:** wind and solar, charging stations for electric vehicles, and pipelines for captured carbon dioxide and hydrogen.
- **Price cap on carbon:** carbon tax
- **Cleaner electricity standards:** A renewable portfolio standard
- **Clean fuel standards:** electric vehicles, enhanced biofuels, electrofuels, and other low-carbon alternatives.
- **Clean product standards:** low-emissions cement, steel, plastics, and other carbon-intensive products.

To sum up,

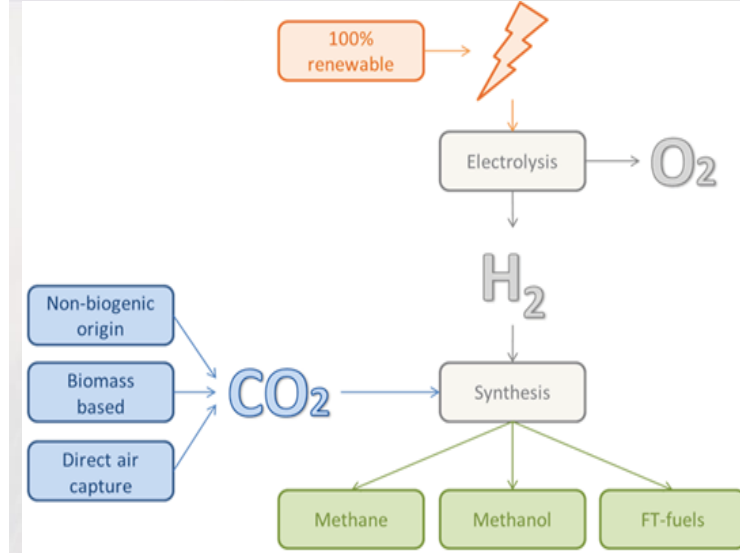
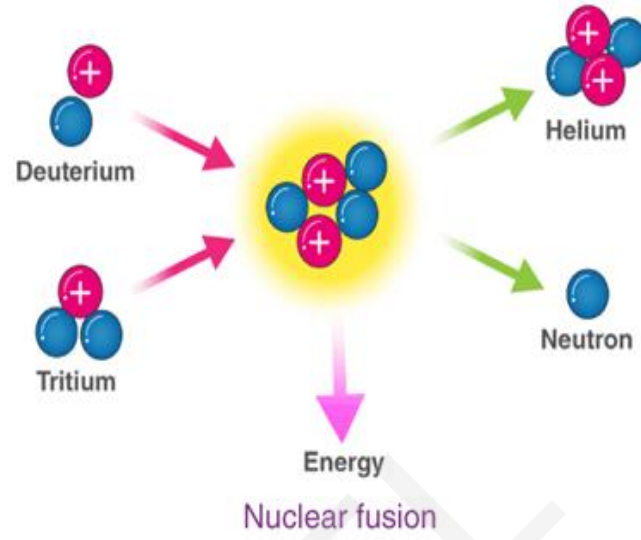
Short term goal: The initial focus should be on the ones with a low Green Premium, or no premium at all.

Long term goal: wherever we decide Green Premiums are too high. That's where the extra cost of going green will keep us from decarbonizing and where there exists a window of opportunity for new technology, businesses, and products to make it more inexpensive.



• **Green Technology Framework: Technological Breakthroughs**

- Production of Hydrogen without carbon emission
- Grid-scale electricity storage with a season's worth of capacity
- Advanced biofuels, Electrofuels
- Zero-carbon cement, Zero-carbon steel
- Plant- and cell-based meat and dairy
- Zero-carbon/Bio fertilizer
- Next-generation nuclear fission
- Nuclear fusion
- Carbon capture (both direct air capture and point capture)
- Underground electricity transmission
- Plastic management
- Drought- and flood-tolerant food crops
- Coolants that don't contain F-gases
- Other non-conventional renewable sources
- Cost-reduction and technical innovation



Why hydrogen?

A versatile, zero-emission, efficient energy carrier

Infinite supply

Easily transported in large volumes

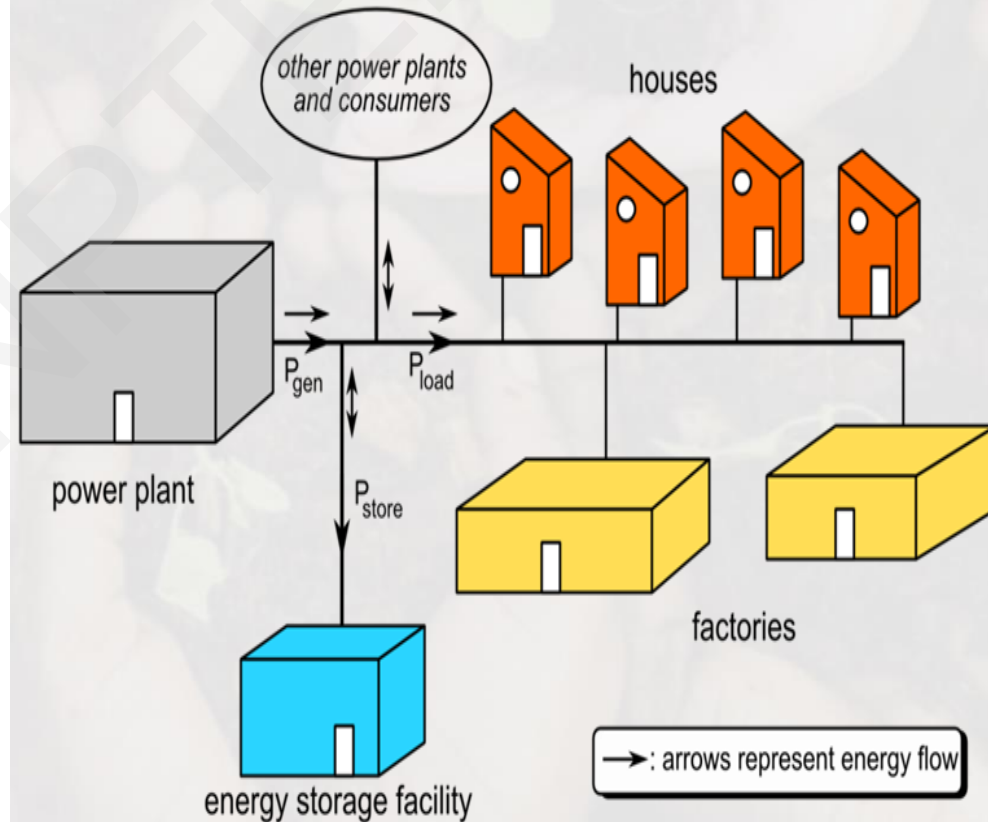
Produces clean power and/or heat at the point of use



Can be produced with no carbon footprint from multiple sources

High energy density

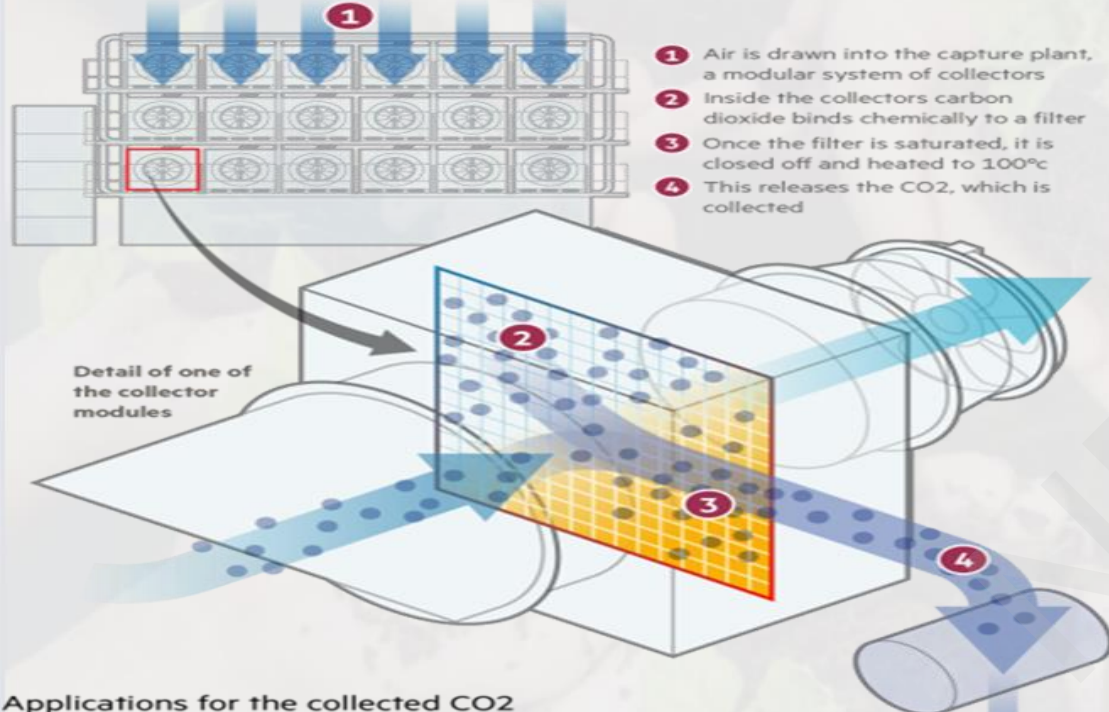
Can be stored in large quantities and for long periods





What is direct air capture?

A method for removing the excess CO₂, Climeworks's system works as follows



10 Key Solutions Needed to Reduce Greenhouse Gas Emissions

- 1.**  **PHASE OUT** coal plants
- 2.**  **INVEST** in clean energy & efficiency
- 3.**  **RETROFIT** buildings
- 4.**  **DECARBONIZE** cement, steel & plastics
- 5.**  **SHIFT** to electric vehicles
- 6.**  **INCREASE** public transport
- 7.**  **DECARBONIZE** aviation and shipping
- 8.**  **HALT** deforestation & **RESTORE** degraded lands
- 9.**  **REDUCE** food loss and waste
- 10.**  **EAT** more plants & less meat

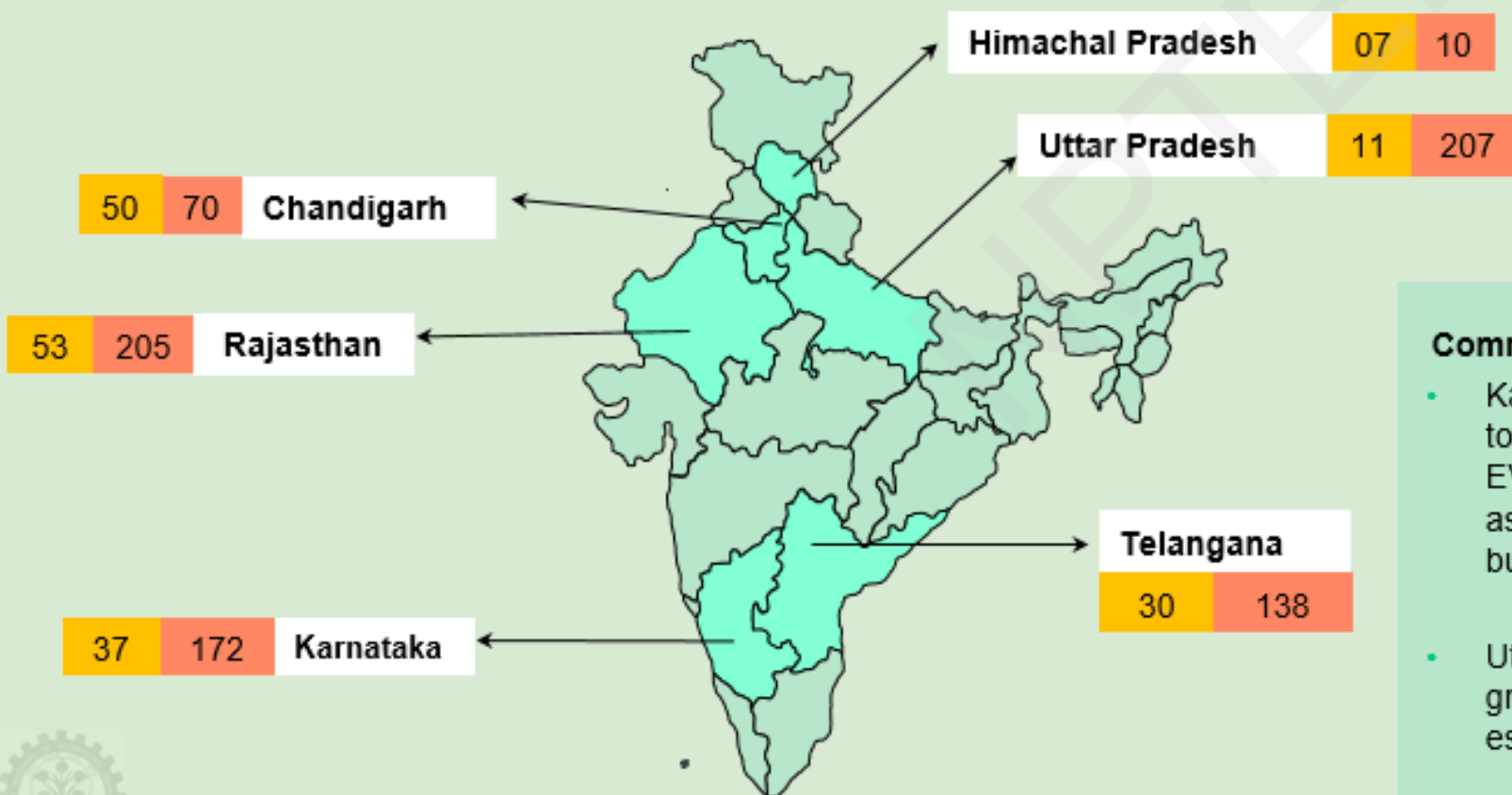
Source: WRI.

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Currently, as per EV industry body - Society of Manufacturers of Electric Vehicles - there are **1,800 charging stations in India as of March 2021** for approximately 16,200 electric cars, including the fleet segment.

Charging Stations established under FAME Schemes



FAME I Scheme

FAME II Scheme

Comment

- Karnataka was the first state to introduce a comprehensive EV policy and has emerged as a hotspot for EV businesses in India
- Uttar Pradesh has the fastest growth in EV charging establishment today



➤ **Judicial decisions highlighting climate action**

• **INTERNATIONAL CASES ON CLIMATE ACTION**

- The State of the Netherlands v. Stichting Urgenda
- Asghar Leghari v. Federation of Pakistan, 2015
- Friends of Irish Government CLG v. The Government of Ireland and the Attorney General
- Rddhima Pandey v. UOI and the Pollution Control Board
- Juliana v. United States



- **Legality of Instruments**
- **UNFCCC Preamble**
- Recognizing that States should enact effective environmental legislation, that environmental standards...
- Affirming that responses to climate change should be coordinated with social and economic development in an integrated manner with a view to avoiding adverse impacts...
- **UNFCCC principles**
- Art 3
- 1. The Parties should protect the climate system for the benefit of present and future generations of humankind, on the basis of equity and in accordance with their common but differentiated responsibilities and respective capabilities. Accordingly, the developed country Parties should take the lead in combating climate change and the adverse effects thereof.



contd...

- 3. The Parties should take precautionary measures to anticipate, prevent or minimize the causes of climate change and mitigate its adverse effects... Where there are threats of serious or irreversible damage, lack of full scientific certainty should not be used as a reason for postponing such measures...
- 4...should be integrated with national development programme/s,...
- Art 4
- 1. All Parties, taking into account their common but differentiated responsibilities and their specific national and regional development priorities, objectives and circumstances, shall:...



- **Judicial decisions highlighting climate action(Indian Scenario)**

- Legislative Provisions:

- Article 51-A
- Article 32 & 226
- Article 19
- Article 21

- **Indian Judiciary's Perspective:**

- M.C. Mehta v Kamal Nath [(1997) 1 SCC 3880]
- M.C Mehta v Kamal Nath [AIR 2000 SC 1997]
- Vellore citizens welfare forum v. UOI, AIR 1996 SC 2715
- Indian Council for Enviro Legal Action v. UOI, AIR 1996 SC 1446
- Principles formulated:
 - Public Trust Doctrine
 - Polluter Pays Principle



- **PARIS AGREEMENT**

- The Paris Agreement is a legally binding international treaty on climate change. It was adopted by 196 Parties at COP 21 in Paris, on 12 December 2015 and entered into force on 4 November 2016.
- The Agreement sets **long-term goals to guide all nations:**
 - substantially reduce global greenhouse gas emissions to limit the global temperature increase in this century to 2 degrees Celsius, and preferably to 1.5 degrees Celsius, compared to pre-industrial levels.
 - **review countries' commitments every five years;**
 - provide financing to developing countries to mitigate climate change, strengthen resilience and enhance abilities to adapt to climate impacts.
- The Paris Agreement provides a durable framework guiding the global effort for decades to come. It marks the beginning of a **shift towards a net-zero emissions world.**
- It is a **landmark in the multilateral climate change process** because, for the first time, a binding agreement brings all nations into a common cause to undertake ambitious efforts to **combat climate change and adapt to its effects.**

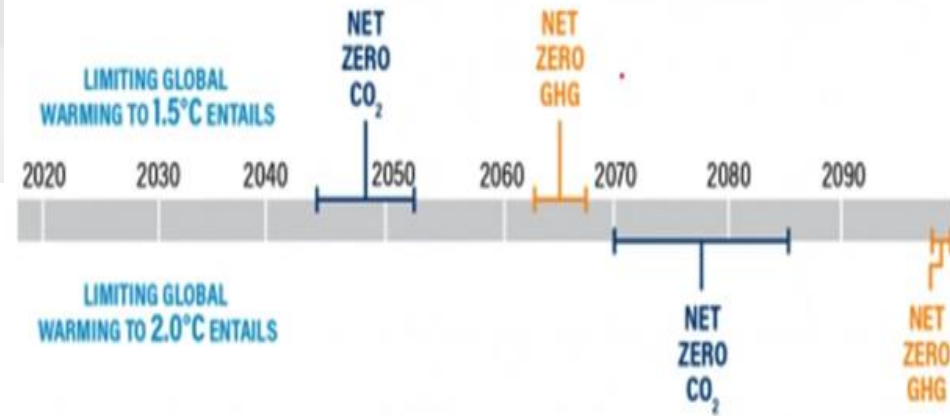


contd...

- Following announcements are expected:
- Rules still needed to implement the Paris Agreement
- Making a faster switch to electric cars
- Speeding up the phasing out of coal power
- Cutting down fewer trees
- Protecting more people from the impacts of climate change, such as funding coastal-defence systems.
- Up to 25,000 people are expected in Glasgow, including world leaders, negotiators and journalists.
- Tens of thousands of campaigners and businesses will also be there to hold events, network - and hold protests. Extinction Rebellion, for example, are calling for an immediate end to the use of fossil fuels.
- At the end of the conference, some form of declaration is expected.
- Every country will be required to sign up and it could include specific commitments.
- Russia and China decided to skip the COP26 climate summit, even though China and Russia together account for 32% of global CO2 emissions!



Global timeline to reach net-zero emissions



Source: IPCC Special Report on Global Warming of 1.5°C

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COP26 in numbers

The 26th 'Conference of the Parties' climate summit is being held in Glasgow from 31 October to 12 November

Who will be there?

25,000

delegates from
200 countries

10,000

police officers from
across the UK

100,000

people expected
to protest



What are the targets?

1.5°C

is the agreed limit
for the temperature
to rise to avoid
climate catastrophe

Nearly 200 countries

are being asked to
submit new plans
to reduce their
emissions

\$100bn

is what wealthy
countries are being
asked to give to
help poorer nations



TRACKING PROGRESS

With the Paris Agreement, countries established an enhanced transparency framework (ETF). Under ETF, starting in 2024, countries will report transparently on actions taken and progress in climate change mitigation, adaptation measures and support provided or received. It also provides for international procedures for the review of the submitted reports.

Information collection and preparation-

The information gathered through the ETF will feed into the Global stocktake which will assess the collective progress towards the long-term climate goals. At the COP24 meeting in Katowice, Poland, in 2019, they agreed on a three-phase process for the stocktake.



World on track for
2.7°C temperature rise
this century, warns UN



- **Communicating and acting on findings.**
- Although climate change action needs to be massively increased to achieve the goals of the Paris Agreement, the years since its entry into force have already sparked **low-carbon solutions and new markets**. More and more countries, regions, cities and companies are establishing carbon neutrality targets. **Zero-carbon solutions** are **becoming competitive** across economic sectors representing 25% of emissions. This trend is most noticeable in the power and transport sectors and has created many new business opportunities for early movers.
- By **2030**, zero-carbon solutions could be competitive in sectors representing over 70% of global emissions.



- **No country** performed well enough to achieve the goals of 2015 Paris Agreement
- Six G20 countries are among **"very low" performers**, according to the latest edition of the Climate Change Performance Index (CCPI) released by non-profit Germanwatch, launched in 2005, which tracks the climate protection performance of countries. Rankings are based on their performance under four aspects:
 - **reduction** of greenhouse gas emissions (40 % weightage).
 - use of renewable energy (20 %);
 - energy-use **efficiency** (20 %) and
 - climate **policy** (20 %).
- The **United States (US)**, with a rank of **61**, was the **worst performer**, followed by Saudi Arabia, Iran, Canada and Taipei. **Australia, Russia and South Korea** are the other three major countries with **"very low" ratings**.
- The **top three spots of the ratings are kept empty** because "no country performed well enough". Sweden is ranked fourth, followed by the UK, Denmark, Morocco and Norway.
- **India, at 10**, is the only Asian country with a **"very high" rating**. Egypt, Chile, the UK and Malta ranked the highest under **reducing greenhouse gas emissions** category.
- In the **renewable energy use category**, **Latvia, Norway, Sweden, Denmark and Finland** were the top five countries, **India ranked 27** with a medium rating, Iran, Russia, Taipei, Malaysia and Belarus ranked the lowest. Mexico, Egypt and Canada: were the only three other countries with "very low" rating.



GHG Emissions: Key developments – COVID-19 effects: Global emissions are declining

- Owing to the COVID-19 crisis, the first half of 2020 brought a drastic 8.8% decrease in global GHG emissions. **This is the largest half-year decrease in emissions ever recorded.**
- Till May 2020 there was a **4%-7% worldwide decline** for the year.
- For 1.5°C, world emissions would need to continue to decline by this year's rate.
- **Key results:** GHG Emissions

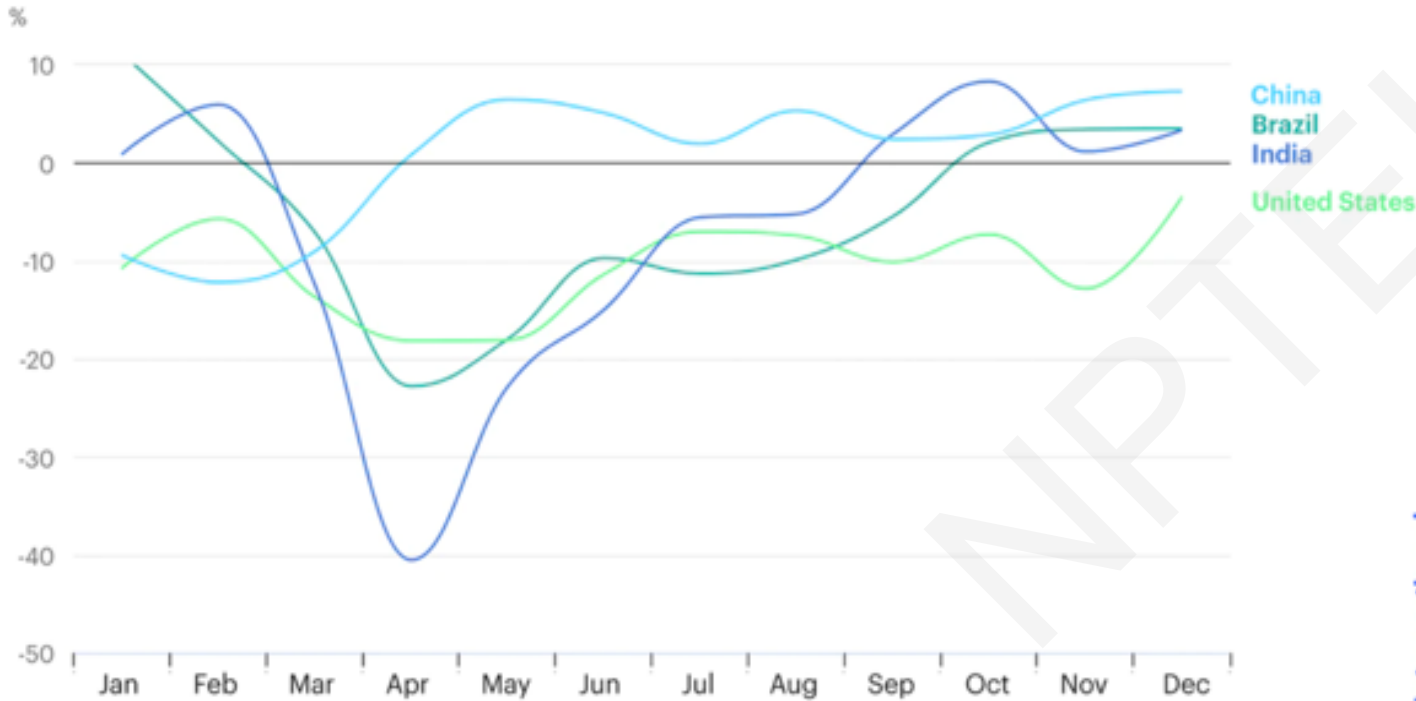
G20 performance:

- No country's performance is rated very high for all indicators in the GHG Emissions category while **only France joins last year's two high performing G20 countries India and the United Kingdom.** Although India has one of the largest growth trends, per capita emissions stay at a comparatively low level, rated very high for their well-below 2°C compatibility.
- Eight of the G20 countries rank as very low performing countries. Saudi Arabia, the worst performing G20 member has moved up from the last to the second last rank.
- EU performance: As last year, the **EU is rated medium for its performance in the GHG Emissions category.** Five EU countries rank as high performers in this year's GHG Emissions rating (excluding the UK). Cyprus, Ireland and Estonia are the worst performing EU countries, all with an overall very low rating in this category.



Monthly evolution of CO₂ emissions in selected major economies, 2020 relative to 2019

Global Energy Review: CO₂ Emissions in 2020



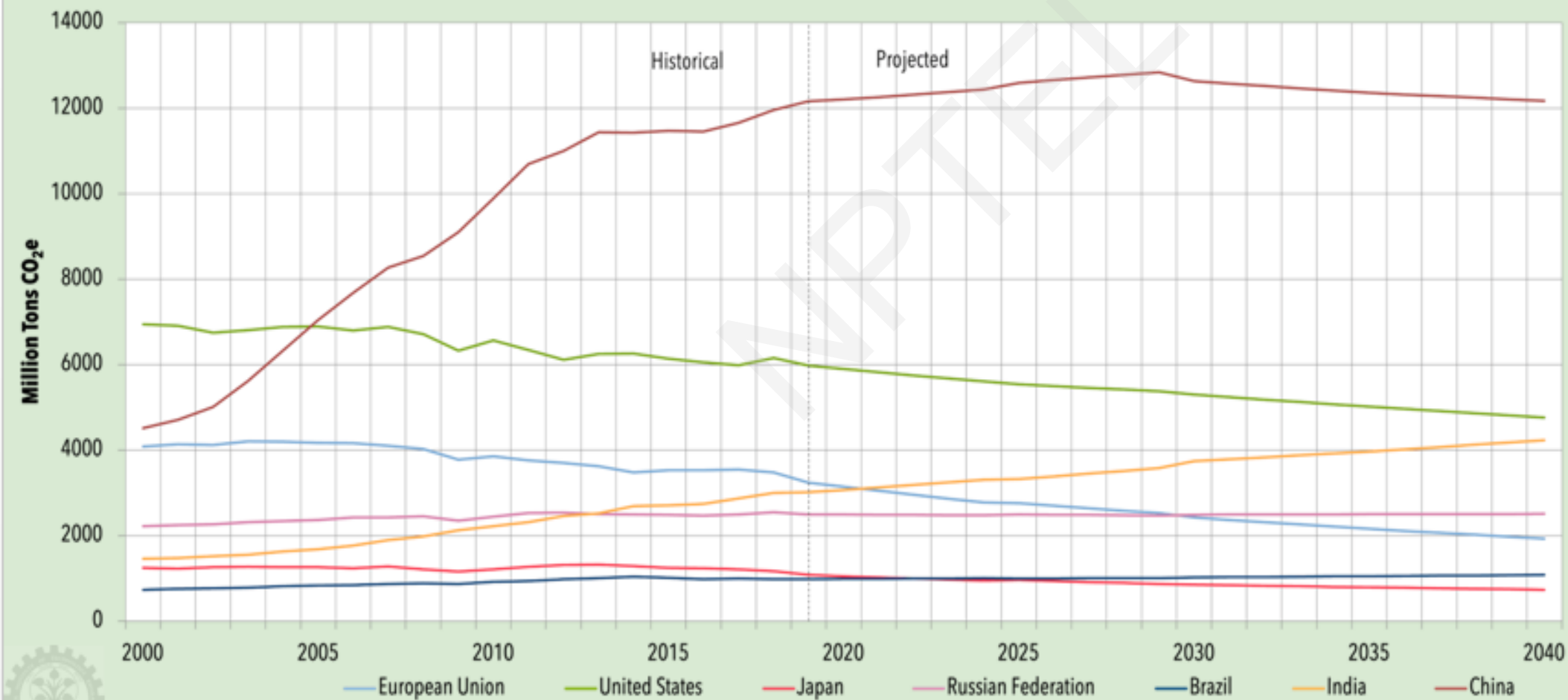
International
Energy Agency

- In 2021 global energy-related CO₂ emissions are projected to rebound and **grow by 4.8% as demand for coal, oil and gas rebounds with the economy**. The increase of over 1 500 Mt CO₂ would be the largest single increase since the carbon-intensive economic recovery from the global financial crisis more than a decade ago, it leaves **global emissions in 2021 around 400 Mt CO₂, or 1.2%, below the 2019 peak**.

- Global CO₂ emissions **declined by 5.8% in 2020**, or almost 2 Gt CO₂ – the largest ever decline and almost five times greater than the 2009 decline that followed the global financial crisis. CO₂ emissions fell further than energy demand in 2020 **owing to the pandemic** hitting demand for oil and coal harder than other energy sources while renewables increased.
- Despite the decline in 2020, **global energy-related CO₂ emissions remained at 31.5 Gt**, which contributed to CO₂ reaching its highest ever average annual concentration in the atmosphere of **412.5 parts per million in 2020** – around **50% higher** than when the industrial revolution began.



Greenhouse Gas Emissions for Major Economies, 1990–2030





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India's Action Plan

(Source: <https://dst.gov.in/climate-change-programme>)

India's Action Plan

The Government of India launched National Action Plan on Climate Change (NAPCC) on 30th June, 2008 outlining eight National Missions on climate change. These include:

1. National Solar Mission
2. National Mission for Enhanced Energy Efficiency
3. National Mission on Sustainable Habitat
4. National Water Mission
5. National Mission for Sustaining the Himalayan Ecosystem
6. National Mission for a Green India
7. National Mission for Sustainable Agriculture
8. National Mission on Strategic Knowledge for Climate Change

The main goals that the NAPCC aims at are:

- **Inclusive** sustainable development strategy
- Achievements of **qualitative changes** to promote national growth
- Adaptation and careful **mitigation** of GHG emissions
- **Regulatory mechanisms** to promote sustainable development
- Invite **international cooperation** for R&D for data transfer and sharing

Suggestions: Robust plans that are flexible in order to accommodate the drastic changes in environment across the country, constant revision of climate change policies

Professor **Subimal Ghosh**, one of the lead authors of the IPCC report and faculty at IIT Bombay said:

"A compound extreme event is when one particular region faces multiple extreme weather events, like the Sundarbans in India. There maybe a time period when, say, a cyclone may have hit the region, which will be immediately followed by high tides that will break embankments and cause salt water to enter agricultural fields. This may again be immediately followed by an intense spell of rainfall"



India's Year of Extremes- Current State

Severely impacted by COVID 19, which has further reduced the **resilience of climate change vulnerable populations already at risk of displacement by storms, floods, droughts** and other climate disasters. With 77% more than normal rainfall recorded in 381 districts of the country out of 683, 265 districts received no rainfall at all. Temperatures were lower by 4-7°C, in parts of Uttar Pradesh, Bihar and Jharkhand. India's current Climate Action Tracker rating is “**HIGHLY INSUFFICIENT**” indicating that **India's climate policies** and commitments are **not consistent with the Paris Agreement's** 1.5°C temperature limit. Under current targets and policies, emissions will continue to rise and are consistent with 4°C or more of warming.





- As a response to the economic crisis **after the pandemic, Indian government unveiled one of the largest stimulus packages** in the world, equating to a share of around 11% of the country's GDP in 2019, which mainly supports activities related to industries and is likely to have a large negative impact on the environment by, for example, increasing the use of fossil fuels, and unsustainable land use.
- The most **recent stimulus (2021)** is more **climate-friendly, with two-thirds of the resources targeted towards a green recovery, including roughly USD 3bn in battery development and solar PV**. While the additional stimulus is a positive step, India continues to support coal, with fresh loans to a number of thermal power projects, undermining a green recovery.
- **Emissions will rise by 2030, due to coal expansion plans**. India's coal capacity would increase from current levels of over 200 GW to almost 266 GW by 2029-2030, with 35 GW expected to come online in the next five years: an increase of 17.5% in coal capacity.



Projects of Climate Change Programme Division (2020-2021)

TITLE OF THE PROJECT	TOTAL AMOUNT ALLOCATED (₹)	DURATION (YEARS)
Climate Change Impacts and Adaptation for a Climate Resilient North East India (Assam)	6,35,65,808	5
Assessment of Impact of Climate Change on the Geo diversity in the Uttarakhand Himalaya for five most disaster prone Districts of Uttarakhand including vulnerability and Risk Assessment: Implication for Sustainable Development and Policy Making	2,06,33,607	3
Establishing/Strengthening the State Climate Change Cell in the State of SIKKIM (Phase-II)	2,72,31,476	5
Strengthening of State Climate Change Cell of Manipur – 2nd phase	2,14,46,160	5
Establishment/strengthening of State Climate Change Cells (Phase II) (Tripura)	1,49,23,756	5
Punjab State Climate Change Knowledge Centre (Phase-II)	2,73,10,095	5
Impact of Climate Change on Water Resources of Arunachal Pradesh	2,46,51,965	3
Long-term ecological monitoring of forest plots in Mizoram, Northeast India	2,44,67,650	3



CLIMATE ACTION TRACKER OVERALL RATING FOR INDIA - HIGHLY INSUFFICIENT

- **POLICIES AND ACTION- ALMOST SUFFICIENT**

- India's current climate policies and actions are not yet consistent with the **Paris Agreement's 1.5°C temperature limit** but could be with moderate improvements, and greenhouse gas emissions are projected to reach a level of 3.84-4.02GtCO_{2e} in 2030.
- India has ambitious renewable energy policies, and has a 2022 renewable energy capacity **target of 175 GW**. As of July 2021, it was just shy of the 100 GW mark, with 98.9 GW of new renewable energy capacity installed, meaning that it is unlikely to reach the target.
- To be aligned with the Paris Agreement 1.5°C limit, India must phase out coal use from its power sector by 2040. The **National Electricity Plan (NEP) in 2018** is projected to add more than 45 GW of coal-fired capacity by 2027, and in 2029-30 India's coal capacity will be increased by 64 GW above 2021 levels, which will result in increase in emissions. .
- Only about **50% of planned coal capacity (6.8 GW)** was installed in 2019–20, and the number of cancelled plants is also increasing, but India still has one of the largest coal pipelines in the world. India would need to move faster towards decarbonising its energy sector.

- **INTERNATIONALLY SUPPORTED TARGET- CRITICALLY INSUFFICIENT**

- India's first NDC has three main elements:
 - An emissions-intensity target of **33%–35% by 2030 below 2005 levels**;
 - To increase the share of non-fossil-based energy resources to **40% of installed electric power capacity by 2030** with support;
 - To create an additional (cumulative) **carbon sink of 2.5–3 GtCO_{2e}** through additional forest and tree cover by 2030.
- India's 40% non-fossil capacity target, as being conditional on international support, needs substantial improvements to be consistent with the Paris Agreement's 1.5°C temperature limit. India has not yet submitted an updated NDC.

- **FAIR SHARE TARGET- HIGHLY INSUFFICIENT**

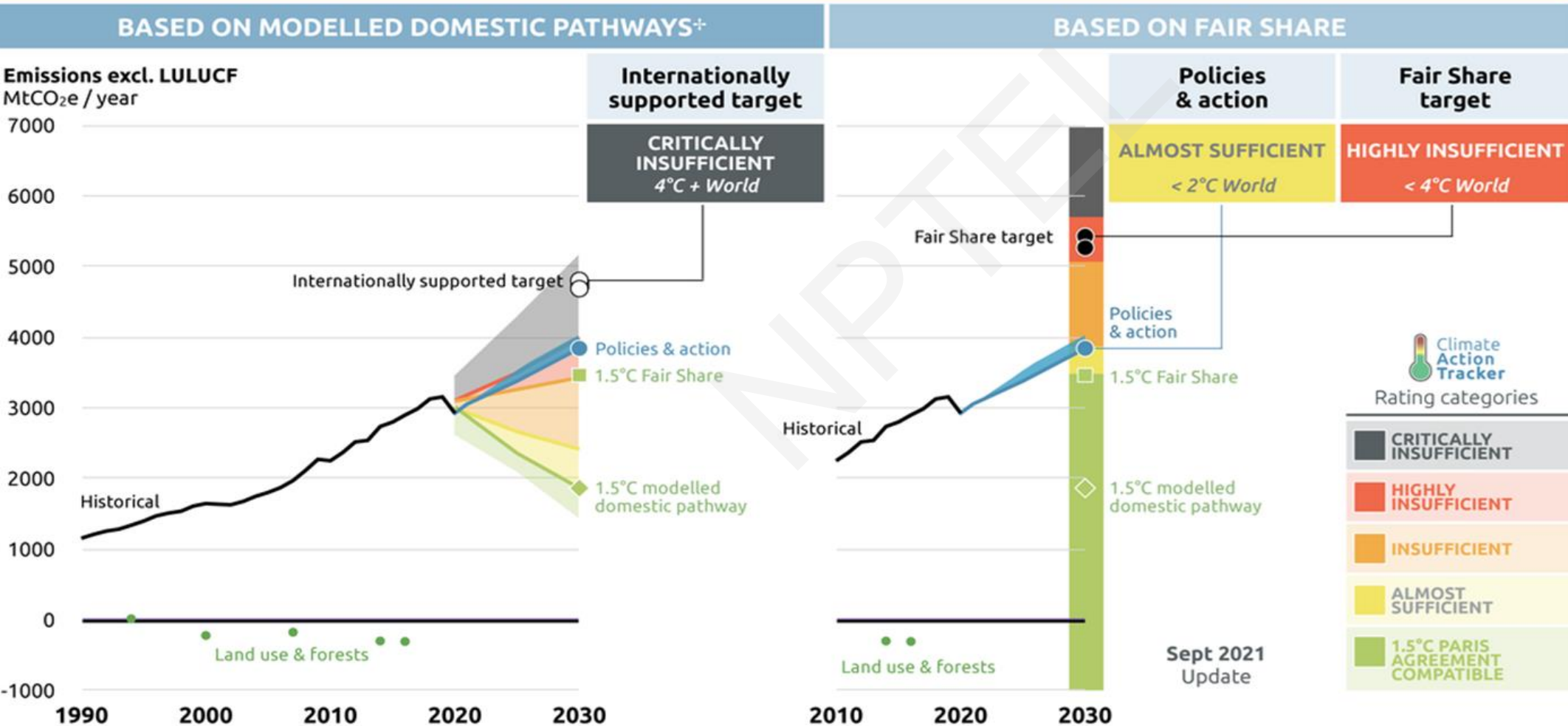
- India's climate policies and commitments are **not consistent** with any interpretation of a fair-share contribution and lead to rising, rather than falling, emissions.

- **NET ZERO TARGET- NO TARGET**

- India does not have a net zero target.



INDIA OVERALL RATING HIGHLY INSUFFICIENT



Rating categories

CRITICALLY INSUFFICIENT

HIGHLY INSUFFICIENT

INSUFFICIENT

ALMOST SUFFICIENT

1.5°C PARIS AGREEMENT COMPATIBLE

* Modelled domestic pathways reflects a global economic efficiency perspective with pathways for different temperature ranges derived from global least-cost models



What ESD Can do????

PEDAGOGICAL FRAMEWORK FOR SUSTAINABLE ENERGY

CURRENT STATUS

Renewable energy as an independent discipline for the undergraduate degree is run in a few technical institutes in India. It is an interdisciplinary field and therefore is taught at a preliminary level in many UG, PG and PhD courses but No IITs or NITs offer a specialized program on renewable energy technology (RET) undergraduate engineering degree program.

POLICY 1

GENERAL AWARENESS: At the time of such a crisis it becomes necessary to target not only the educated few but to spread awareness, teach and train each and every contributing citizen of INDIA.

POLICY 2

EDUCATION POLICIES: Evident from the data, there is a serious lack in Renewable energy courses at technical level. There can be two approaches to increase the involved students:

- with the increase in job opportunities
- through the understanding that learning about sustainable growth is as necessary as learning a language and making is compulsory for all

AFFORDABLE & CLEAN ENERGY



Only a few IITs offer PG programs related to renewable energy technology.

IIT Delhi, Kanpur, Kharagpur and Bombay have an independent energy center for research.

CURRENT STATUS →

IIT Kharagpur offers the dual degree in electrical engineering with specialization in power and energy system.

Four NITs have an independent department or research centre of energy studies.

Indian government has taken multiple initiatives to enhance teaching and learning through the online open web medium.

virtual labs

open source simulation tools

textbook companion project (Scilab codes of mathematical problems)

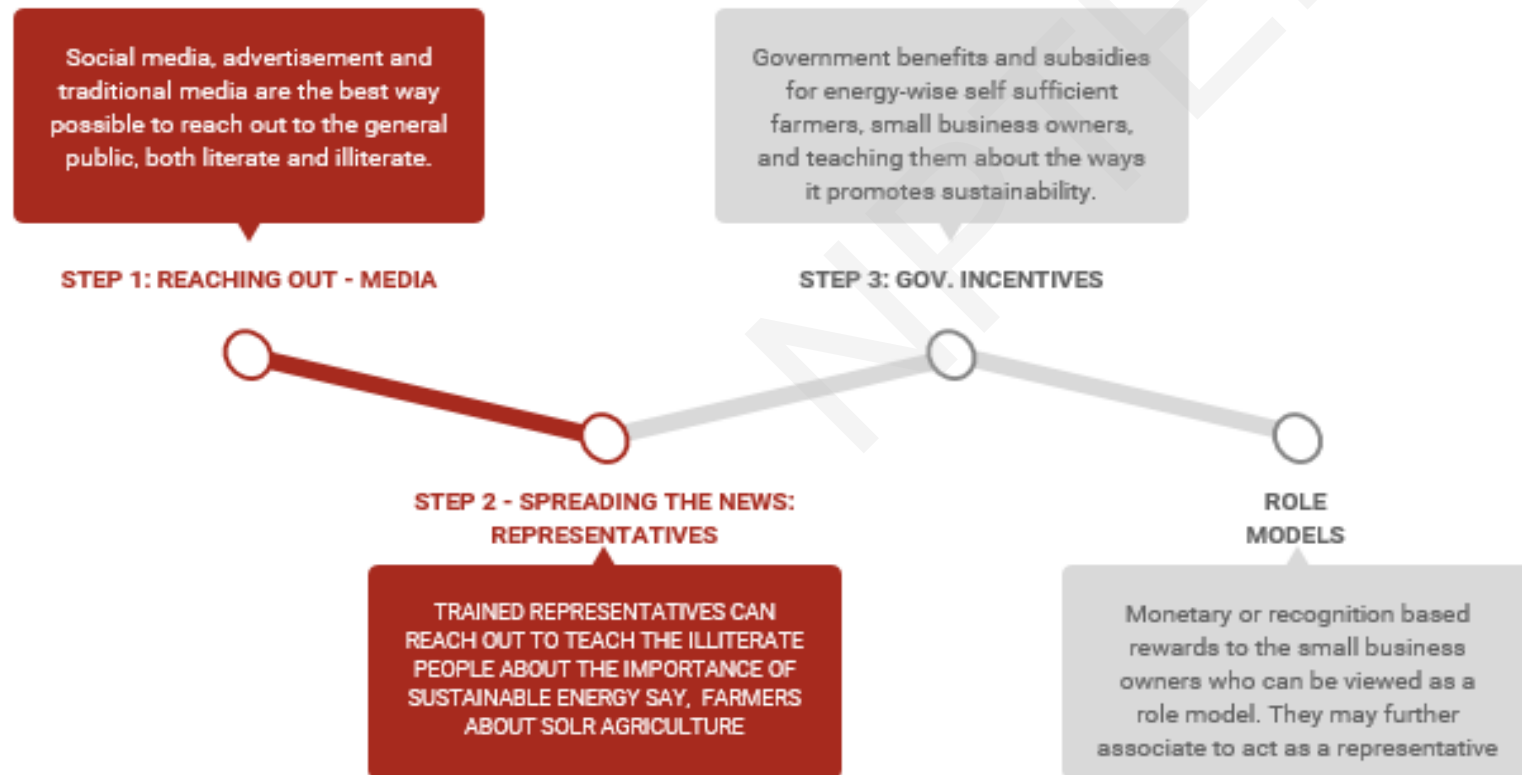
online videos lecture database through National Program for Technology Enhanced Learning(NPTEL)

online thesis management system



POLICY 1 →

GENERAL AWARENESS: At the time of such a crisis it becomes necessary to target not only the educated few but to spread awareness, teach and train each and every contributing citizen of INDIA.



**AFFORDABLE &
CLEAN ENERGY**



LECTURE: 34

Sustainable and Clean Energy



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POLICY 2 →

EDUCATION POLICIES: Evident from the data, there is a serious lack in Renewable energy courses at technical level. There can be two approaches to increase the involved students: with the increase in job opportunities through the understanding the necessity of renewable energy and making it compulsory for all

HIGHER SECONDARY LEVEL

As per NEP 2020, the students get to be trained through vocational training. This can serve as the basis for their introduction to green energy and they can be asked to observe and learn about it. Further, there could be training in implementation of the green energy sources.

GRADUATE LEVEL

Through a two phase process of a) MHRD making the course compulsory and b) improving the job opportunities the study of renewable energy can be promoted among technical institutes.

JOB APPLICATION BASED

Emphasising the importance of clean energy and providing workshops for the employees will go a long way. Special department to ensure the zero carbon footprint and audit the company for the same will not only increase the job opportunities in the field but also make the companies more responsible.



RESPONSIBILITIES AS GLOBAL CITIZEN

POLICY 1

BIOENERGY: It can be the heart of the revolution India and the world is waiting for. Spreading awareness about Bioenergy and implementing it at household level serves two purposes:

- a) Reduces the dependence on grid power and helps cope during this transient phase of shifting towards becoming carbon neutral
- b) Gives every individual a sense of responsibility towards the global crisis

POLICY 2

Solar panels in agricultural practices, coffee shops, small businesses and other public spaces will further the idea of sustenance of affordable and clean energy.



BIOENERGY →

It is essential for a country like India to focus on bioenergy, because of the geographical constraints, and economic constraints. BIOENERGY is not only cheap, affordable, green but it is renewable and self sufficient. Unlike Solar or wind energy which requires large infrastructure. It is implementable

BIOENERGY		OTHER RENEWABLE SOURCES	
INFRASTRUCTURE REQUIREMENTS	LOW	HIGH	SOLAR / HYDROPOWER/ WIND
SELF SUFFICIENCY OR SUSTAINABILITY	HIGH	LOW	REQUIRES CERTAIN FLOW RATE FOR HYDROPOWER. SOLAR PANNELS AND WIND TURBINE PRODUCE ENERGY FOR GIVEN PREIOD IN THE DAY.
TECHNICAL KNOWLEDGE REQUIRED	LOW	HIGH	ENGINEERS ARE REQUIRED TO BE TRAINED TO CONSTRUCT AND OPERATE THEM



HOUSEHOLD POLICIES →



Recently the **Government of the National Capital Territory of Delhi (GNCTD)** decided to utilize its 31 sq. km of rooftop space to farm solar energy. Several incentives like **GBI(Grid based Incentives)** of INR 2.00 per unit (kWh) of gross solar energy generated would be given.

This makes the regular households feel a part of the solution and also helps spread awareness.

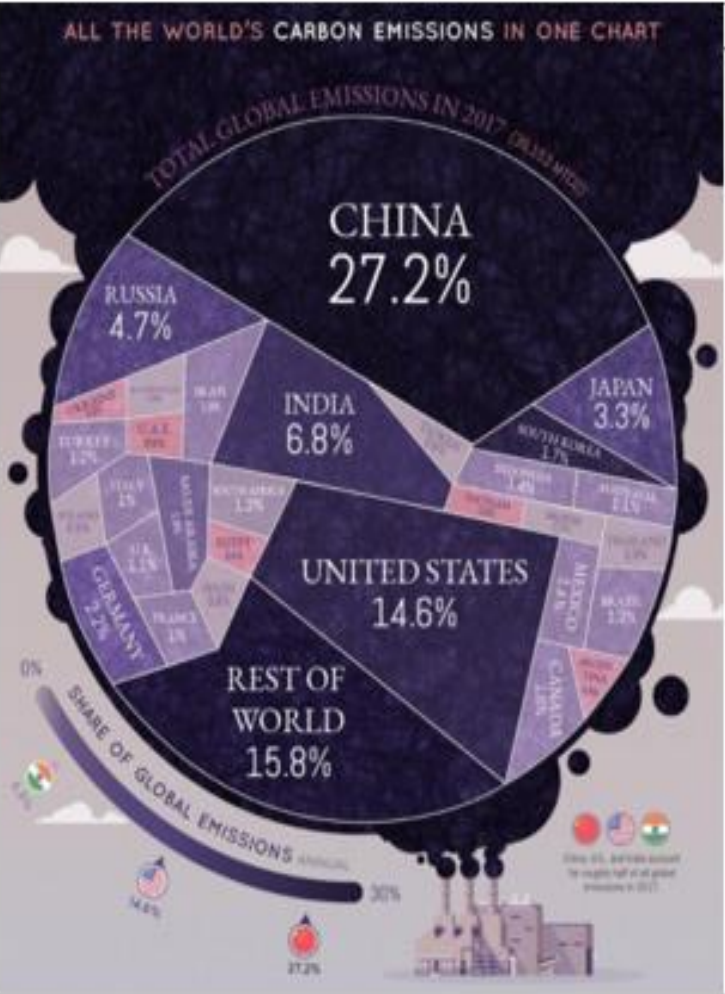
CONCLUSION:

Through the pedagogical framework implementation and an inclusive solution method to affordable and clean energy, India can meet its target of becoming carbon neutral.

This ensures that the measures taken are also followed in the future.



CARBON NEUTRAL: A GOAL TO BE ACHIEVED





INDIA'S PROMISE OF CARBON NEUTRAL

- Recently after the G20 summit in Italy, PM Modi attended the COP26 (Conference of Parties) climate change meeting in Scotland. There he made a promise to transform India into a carbon-neutral country. Though the deadline set by India might seem very late considering that it is 20 years behind that of the US and a decade behind China but unlike countries in the west, and even China, India is decades away from its peak in terms of economic growth and energy consumption, and India's energy demand is expected to grow faster than any other country over the next few years. Its population is also expected to overtake that of China to make it the largest in the world.

INDIA and NSG



- India demanded to be a member of the Nuclear Group **NSG (Nuclear Suppliers Group)** so as to be able to achieve the carbon-neutral goal by the set deadline.
- India made a renewed push for its membership to the coveted **Nuclear Suppliers Group or NSG** ahead of the **COP26 climate summit**, underlining that its climate and development goals are tied with its entry into the grouping.

As of 2020, the NSG has 48 participating governments

•  Argentina	•  Denmark	•  Latvia	•  Serbia
•  Australia	•  Estonia	•  Lithuania	•  Slovakia
•  Austria	•  Finland	•  Luxembourg	•  Slovenia
•  Belarus	•  France	•  Malta	•  South Africa
•  Belgium	•  Germany	•  Mexico	•  South Korea
•  Brazil	•  Greece	•  Netherlands	•  Spain
•  Bulgaria	•  Hungary	•  New Zealand	•  Sweden
•  Canada	•  Iceland	•  Norway	•  Switzerland
•  People's Republic of China	•  Ireland	•  Poland	•  Turkey
•  Croatia	•  Italy	•  Portugal	•  Ukraine
•  Cyprus	•  Japan	•  Romania	•  United Kingdom
•  Czech Republic	•  Kazakhstan	•  Russia	•  United States



What can individuals do?

Major changes need to come from governments and businesses, but scientists say some small changes in our lives can limit our impact on the climate:

- Take fewer flights
- Live car-free or use an electric car
- Buy energy efficient products, such as washing machines, when they need replacing
- Switch from a gas heating system to an **electric heat pump**
- Insulate your home.



Indian lawmaker submits private bill to achieve net zero emissions by 2050

Published on 18/03/2021, 4:35pm

A lawmaker from India's ruling party has proposed a bill to "start a discussion" on net zero emissions - while defending coal mining in his district





Behavioural Change!

Small efforts big change

- Swachh bharat ['shauch' (bathroom) se 'soch' (mentality/behaviour) tak]]
- COVID vaccine fear mongering initially>PM vaccinated> more ministry awareness camps> publicity> rcaller tunes> **100 crore target!**
- Near my house illustration- vendor sells ice cream and *aaloo chaat* in leaf cup with bamboo spoon (cost effective+zero plastic)
- Recycled footwear, clothes of plastic (spinning process); jute bag; full ban on low micron plastics





Stubble Burning?



Intentionally setting fire to the straw stubble that remains after grains, such as rice and wheat, have been harvested (months of oct -nov bw kharif and rabi)

Why stubble burning is a big challenge?

- It releases huge amount of pollutants and gasses like NO_2 , N_2O , CO_2 , SO_2 , CH_4 etc
- excessive heat kills the soil microbes, reduces soil moisture as well as destroys the nutrients.
- It reduces the visibility and thus affects the free vehicular movement.
orth westerly winds carry the pollutants towards Delhi.
- respiratory and cardiovascular diseases owe their origin to environmental pollution.

How boomer spray technology can help?

It sprays the bio decomposers on freshly harvested rice field and thus accelerates the process of turning rice stubbles into compost within a short span of 8 days (ordinarily 2 months for decomposition earlier hence farmers resorted to burning)

Bengaluru based firm is providing technology to aid farmers spray decomposers over 5 lakh acres of land



CONCLUSION

- No one is safe till everyone
- Global citizenship
- Take IPCC seriously
- Fix the issues above stated-accountable; review ombudsman
- Global climate fund mandate realization of 100 bn\$ + CBDR (consider historic emitters)
- NZE to be part of customary into law as 193 members of COP. non compliance deterrent
- Net Zero to Net minus is India's stand for developed (and developed want India to commit to net zero).
- goals to commitments should be emphasized
- Bring in Energy Security;
- Increase the share of clean power;
- Increase Energy Availability and Access;
- Improve Energy Affordability; and
- Maximize Energy Equity



2.3 PARIS AGREEMENT

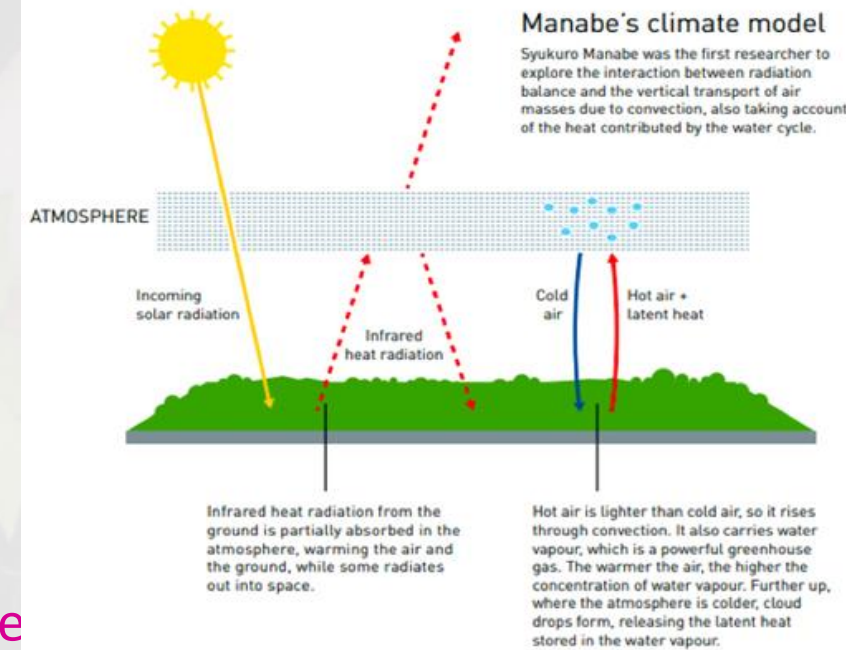
- **Adaptation (Art. 7)**-All Parties should engage in adaptation, including by formulating and implementing **National Adaptation Plans**, and should submit and periodically update an adaptation communication describing their priorities, needs, plans and actions. The adaptation efforts of developing countries should be recognized.
- **Mitigation (Art. 4)**: The Paris Agreement establishes binding commitments by all Parties to prepare, communicate and maintain a **nationally determined contribution (NDC)** and to pursue domestic measures to achieve them.



Nobel Prize!!

For Climate Action..

- The 2021 Nobel Prize in Physics is awarded with one half jointly to Syukuro Manabe, Klaus Hasselmann and the other half to Giorgio Parisi “for groundbreaking contributions to our understanding of complex physical systems.”
- This is the first time climate scientists (Manabe and Hasselmann) have been awarded the Physics Nobel. Manabe and Hasselmann:
- Awarded for work in physical modelling of Earth’s climate, quantifying variability and reliably predicting global warming.
- Demonstrated how increases in the amount of carbon dioxide in the atmosphere would increase global temperatures, laying the foundations for current climate models.
- Parisi: Awarded for “the discovery of the interplay of disorder and fluctuations in physical systems from atomic to planetary scales.”





Challenges

ANTHROPOGENIC GREENHOUSE GAS EMISSIONS have increased since the pre-industrial era, driven largely by **ECONOMIC AND POPULATION GROWTH**, and are now higher than ever.



Solutions

Inform and inspire people and institutions to **TAKE ACTIONS AGAINST CLIMATE CHANGE**



THE WORLD IS WARMING at an alarming rate, damaging our ability to grow food



Innovate and adopt technological solutions to **MITIGATE AND ADAPT TO CLIMATE CHANGE**



STORMS, FLOODS, HURRICANES AND DROUGHTS are intensifying, oceans are warming and becoming more acid, ice is disappearing and seas are rising.



PROTECT THE MOST VULNERABLE against climate change impacts





References

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- <https://www.bing.com/videos/search?q=Sustainable+%26+clean+energy&ru=%2Fvideos%2Fsearch%3fq%3dSustainable%2b%2526%2bclean%2benergy%26FORM%3dHDRSC3&view=detail&mid=3B33894BA98053B63C233B33894BA98053B63C23&&FORM=VDRVSR>
- <https://www.bing.com/videos/search?q=Sustainable+%26+clean+energy&ru=%2Fvideos%2Fsearch%3fq%3dSustainable%2b%2526%2bclean%2benergy%26FORM%3dHDRSC3&view=detail&mid=2F91795AF6B11A37ECFF2F91795AF6B11A37ECFF&&FORM=VDRVSR>
- <https://www.bing.com/videos/search?q=Sustainable+%26+clean+energy&ru=%2Fvideos%2Fsearch%3fq%3dSustainable%2b%2526%2bclean%2benergy%26FORM%3dHDRSC3&view=detail&mid=F42062683B02020AC2B5F42062683B02020AC2B5&&FORM=VDRVSR>
- <https://www.bing.com/videos/search?q=Sustainable+%26+clean+energy&ru=%2Fvideos%2Fsearch%3fq%3dSustainable%2b%2526%2bclean%2benergy%26FORM%3dHDRSC3&view=detail&mid=F3AB8B2E03FEC5B3E534F3AB8B2E03FEC5B3E534&&FORM=VDRVRV>
- <https://www.bing.com/videos/search?q=Sustainable+%26+clean+energy&ru=%2Fvideos%2Fsearch%3fq%3dSustainable%2b%2526%2bclean%2benergy%26FORM%3dHDRSC3&view=detail&mid=BDE5E8200AAB866D466CBDE5E8200AAB866D466C&&FORM=VDRVRV>
- <https://www.bing.com/videos/search?q=Sustainable+%26+clean+energy&ru=%2Fvideos%2Fsearch%3fq%3dSustainable%2b%2526%2bclean%2benergy%26FORM%3dHDRSC3&view=detail&mid=018E9DC9E6797FC44225018E9DC9E6797FC44225&&FORM=VDRVRV>
- <https://www.bing.com/videos/search?q=Sustainable+%26+clean+energy&ru=%2Fvideos%2Fsearch%3fq%3dSustainable%2b%2526%2bclean%2benergy%26FORM%3dHDRSC3&view=detail&mid=17C0C542568E1745774D17C0C542568E1745774D&&FORM=VDRVRV>



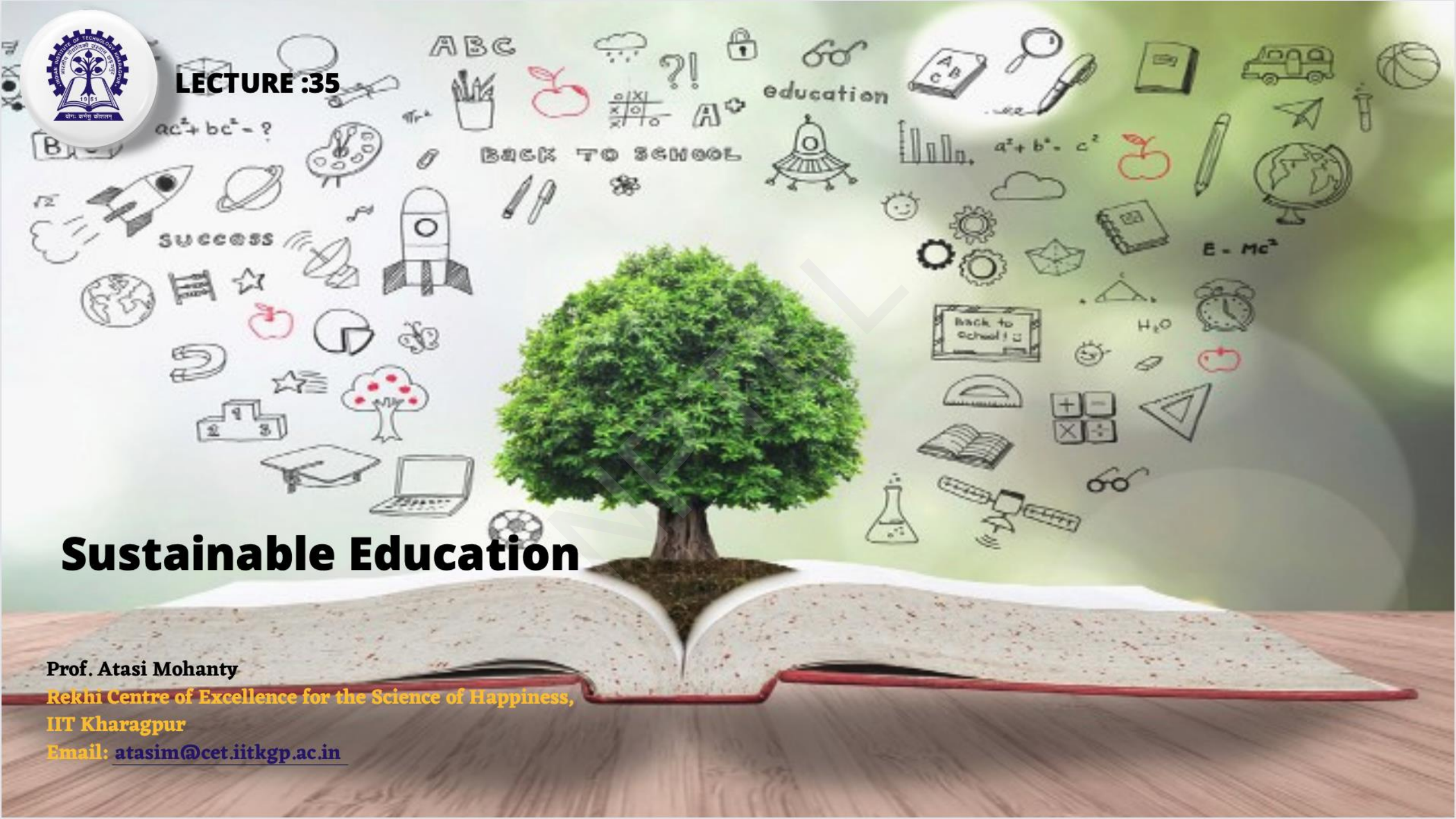
LECTURE :35

Sustainable Education

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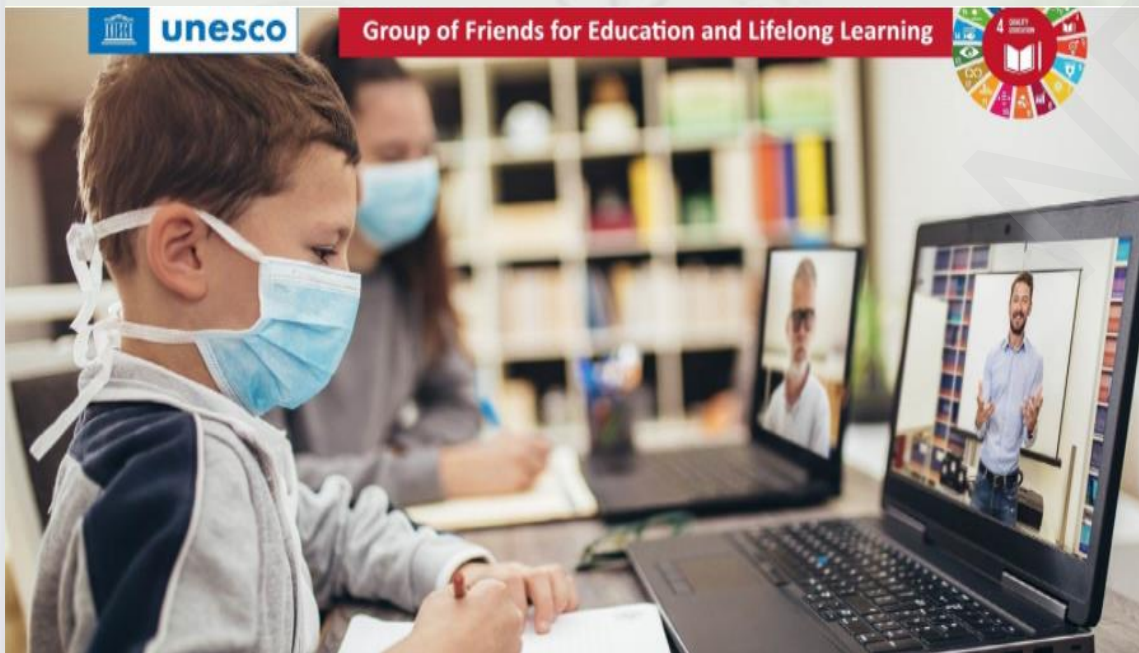
- **United Nations alert that education should be a clear priority**





HESI highlights the role of higher education in building a better world for current and future generations

On 20 October 2020, the Higher Education Sustainability Initiative (HESI) raises the flag of higher education as a driver for sustainable development and inclusive societies on the first day of the Global Education Meeting 2020.





<https://en.unesco.org/vef/load/35048342f3deb9bd63a05fdc1bde118?width=696px&height=357>

4 **QUALITY
EDUCATION**



- **Sustainable Development Goal 4** is the education-related goal of the United National 2030 Agenda for Sustainable Development, adopted in September 2015. Its overall aim is: to: “**ensure inclusive and equitable quality education and promote lifelong learning opportunities for all**”.
- **Education 2030 Framework for Action-**
- **Seven Outcome Targets-**
- **4.1 Universal primary and secondary education-** By 2030, ensure that all girls and boys complete free, **equitable and quality** primary and secondary **education** leading to relevant and effective learning outcomes;
- **4.2 Early childhood development and universal pre-primary education-** By 2030, ensure that all girls and boys have **access to quality early childhood** development, care and pre-primary education so that they are ready for primary education



- **4.3 Equal success to technical/vocational and higher education-** By 2030, ensure equal access for all women and men to affordable and quality technical, vocational and tertiary education, including university
- **4.4 Relevant skills for decent work-** By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship
- **4.5 Gender equality and inclusion-** By 2030, eliminate gender disparities in education and ensure equal access to all levels of education and vocational training for the vulnerable, including persons with disabilities, indigenous peoples and children in vulnerable situations;
- **4.6 Universal youth literacy-** By 2030, ensure that all youth and a substantial proportion of adults, both men and women, achieve literacy and numeracy
- **4.7 Citizenship education for sustainable development -**By 2030, ensure that all learners acquire the knowledge and skills needed to promote sustainable development, including, among others, through education for sustainable development and sustainable lifestyles, human rights, gender equality, promotion of a culture of peace and non-violence, global citizenship and appreciation of cultural diversity and of culture's contribution to sustainable development.



- **Three means of implementation**
- **4.a Effective learning environments-** Build and upgrade education facilities that are child, disability and gender sensitive and provide safe, non-violent, inclusive and effective learning environments for all
- **4.b Scholarships** -By 2030, substantially expand globally the number of scholarships available to developing countries, in particular least developed countries, small island developing States and African countries, for enrolment in higher education, including vocational training and information and communications technology, technical, engineering and scientific programs, in developed countries and other developing countries
- **4.c Teachers and educators-** By 2030, substantially increase the supply of qualified teachers, including through international cooperation for teacher training in developing countries, especially least developed countries and small island developing States



A. One-page summary of *ESD for 2030* framework

EDUCATION FOR SUSTAINABLE DEVELOPMENT

Towards achieving the SDGs: *ESD for 2030*

**GLOBAL ACTION
PROGRAMME**
(2015-2019)

ESD for 2030
(2020-2030)

Vision

ESD for 2030 aims to build a more just and sustainable world through strengthening ESD and contributing to the achievement of all 17 Sustainable Development Goals

Priority action areas

- 2) Transforming learning environments
- 3) Building capacities of educators
- 4) Empowering and mobilizing youth
- 5) Accelerating local level actions

Strategic objective

Promote ESD as a key element of quality education and a key enabler of all 17 Sustainable Development Goals with special attention to a) individual transformation, b) societal transformation & c) technological advances

Target groups

Policy-makers, institutional leaders, learners, parents, educators, youth & communities

ACTIVITIES BY MEMBER STATES: COUNTRY LEVEL MULTI-STAKEHOLDER INITIATIVE

1



Advancing policy

Integrating ESD in global, regional & national policies of education & sustainable development

2



Learning Environment

Promoting whole-institution approach

3



Educators

Providing capacity development opportunities

4



Youth

Providing opportunities for youth engagement

5



Community

Empowering local communities as “nodal” platforms for all priority action areas

Country level multi-stakeholder initiative on ESD

Normative guidelines & country-level support

Inclusive network of partners

Provide regular platforms to meet & exchange

Monitoring & evaluation

Monitor achievements in the 5 priority action areas & programme implementation, in line with SDG Target 4.7 monitoring

Communicating for action

Develop a dedicated programme to inform about the SDGs & the role of education

Evidence-informed implementation & resource mobilization

Monitor emerging issues & trends and mobilize resources to support ESD implementation

UNESCO SUPPORT TO MEMBER STATES